Multi-Agent Reproducibility Auditing with Verifiable Evidence

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Abstract

Reproducibility audits must establish whether published conclusions follow from the underlying data and analysis. We present a multi-agent protocol that links audit findings to source evidence, recorded decisions and human-verifiable checks. Five area auditors examine an article and its deposited materials; author-provided code is optional. A coordinator adjudicates findings, and critics from a second model family challenge the report before gated verification and release. The framework retains confirmed findings, tested negatives and exclusions, making the audit itself inspectable and contestable. An implemented single-agent baseline uses the same sources and evidence standards. Eighteen evaluation quantities assess finding validity, self-correction, compliance, reproducibility and resource use.

1 Introduction

Reproducing a number is one part of checking a scientific result. A published estimate can be recovered exactly even when a coding choice, data-construction error or unsupported interpretation undermines the conclusion. Reproducibility audits therefore connect the article's claims to the underlying data and analysis, investigate benign explanations for apparent discrepancies, and assess the consequences of supported findings. Their own conclusions require evidence that another investigator can inspect and challenge.

Computational reproduction remains difficult even with deposited materials (Chang and Li, 2022; Hardwicke et al., 2018; Herbert et al., 2024; Brodeur et al., 2026a). Language-agent research addresses several parts of this problem: manuscript feedback (Liang et al., 2024), computational reproduction (Siegel et al., 2025; Xu and Yang, 2026), and reproduction, improvement and extension of empirical analyses (Schwartz et al., 2026). Human–AI team studies also examine research-reproducibility assessment (Brodeur et al., 2026b). These contributions make execution and review more accessible. Our focus is how an audit substantiates a finding, resolves challenges to it, and preserves the evidence behind the released report.

We present a protocol in which five area auditors inspect the sources, a coordinator records and adjudicates their findings, and critics from a second model family challenge the audit report. Deposited data accompanying an article are sufficient; the workflow also audits studies without author-provided code by reconstructing supported analyses and checking data and reporting directly. One researcher can operate a single-paper audit, while the implementation also supports coordinated work across a collection of papers.

Contributions. First, a claim-level registry retains source evidence, tested negatives, exclusions and decision histories, linking discovery to the final report. Second, review and release contracts connect cross-family critique, subsequent review of revisions, detector controls and executable verification. This combination makes errors in the audit itself traceable through correction and handoff. Third, an implemented single-agent baseline preserves the task and evidence standards while removing role separation and external critique; a common evaluation design measures finding validity, recovery, compliance and resources. The contribution is this integrated audit protocol and its inspectable artifacts, with a versioned implementation of instructions, runners, schemas and checks.

2 Background and Related Work

Here, *reproduction* uses original data and specifications; replication tests findings with new data

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(Open Science Collaboration, 2015; Camerer et al., 2016).

Automated reproduction and reanalysis. CORE-Bench evaluates agents on computational reproduction of published research (Siegel et al., 2025). Xu and Yang (2026) automate material retrieval, environment reconstruction, execution and output matching, with standardized instrumental-variable diagnostics as a reanalysis application. The NBER workflow of Schwartz et al. (2026) reproduces, improves and extends empirical economics studies. Thus, execution, reanalysis and multi-stage agent workflows are established ingredients. Our contribution specifies how broad audit findings are admitted, challenged, revised and tied to a release: evidence-backed claims about data integrity, methods, reporting and disclosure, including cases in which printed results reproduce.

Manuscript review. LLM feedback tools assess research manuscripts (Liang et al., 2024). Refine.ink provides technical manuscript checks, including parallel cross-examination by multiple models. The American Economic Association describes its use around revise-and-resubmit decisions; Refine also reports a partnership with the Econometric Society (American Economic Association, 2026; Golub, 2026; Refine, 2026). Such checks complement the deposit-grounded audit developed here: tracing published claims through data and code when available, testing remedies, and delivering executable or explicit manual verification. Neither task replaces an editor's assessment of a paper's contribution.

Checks and critique. GRIM, GRIMMER, SPRITE and statcheck test restricted statistical consistency conditions (Brown and Heathers, 2017; Anaya, 2016; Heathers et al., 2018; Nuijten et al., 2016; Nuijten and Wicherts, 2024); package audits can also investigate failures such as leakage (Kapoor and Narayanan, 2023). Self-feedback (Madaan et al., 2023) and multi-agent debate (Du et al., 2024) already use criticism to revise model outputs. We build on these ideas with source-bound findings, retained opposing evidence, adjudication histories and checks connecting the reviewed report to its released artifacts. Separate model families provide another route for challenge; their agreement is not itself evidence that a claim is true.

3 The Protocol

Units and responsibility. A *registry row* records one candidate finding or tested negative; a *verdict* is a critic's judgment on a claim. A *paper-defect* concerns the original research, whereas a *comment-defect* concerns the audit report. These units are analyzed separately. For evaluation, *headline quantities* support the abstract, numbered conclusions or final discussion paragraph; the audit checklist requires reproduction of reported headline values at printed precision.

Five auditors, a coordinator and any post-fan-out verifier use model family A; critics and any targeted adjudicator use family B (Section 4). Independence means separate executions and evidence records, not statistically independent errors. A human operator initiates and oversees the audit and records interventions. The comment identifies its human authors and correspondence contact. The coordinating institution receives reports and handles verification assignments and external correspondence; the released workflow uses the Institute for Replication (I4R). Figure 1 summarizes the stages.

3.1 Stage 0: intake

Intake checks access to the pinned critic model, creates a per-paper workspace, and records the protocol version, environment, package manifest and policy. Deposited inputs remain read-only; computations that write beside inputs use working copies. Data are sufficient package contents: author code is used when available, while missing code is recorded and supported computations are reconstructed from the article and data. A check whose specification cannot be recovered is marked unverifiable. Per-paper workspaces permit concurrent audits of different papers; an advisory reservation records who is handling each paper in a shared collection (Appendix B).

3.2 Stage 1: independent area audits

Five auditors examine reproduction, data integrity, text–data consistency, code and methods, and privacy and archive hygiene. Their checklists overlap (Appendix A). This partition gives each area a dedicated context, tool budget and evidence record, with overlap providing alternative paths to shared risks. It also limits dependence on one session's early interpretation. The count follows the five-area checklist; it is a coverage choice, not an empirically optimized team size. The single-agent baseline cov-

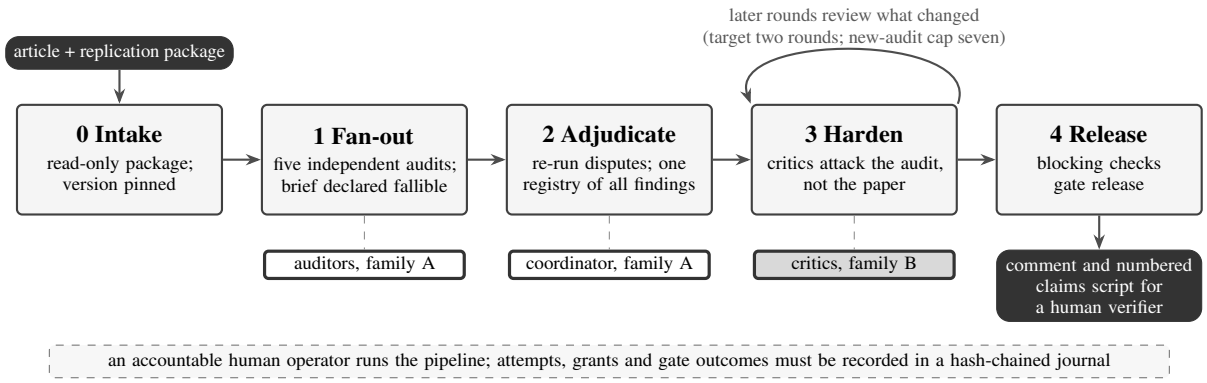


Figure 1: Overview of the intended workflow. Five stages organize a completed audit. Critique repeats for at least two rounds, each later round reviewing what the previous one changed. The architecture separates roles from model identity and requires the critics' model family (B) to differ from the coordinator's and auditors' (A). A fact counts as settled only after confirmations from both families.

ers all five areas in one session.

The skill maintainers supply a shared brief, copied by the runner into each role packet. It defines source locations, evidence and disclosure rules, confidence labels, benign-explanation checks and report structure. It is a common instruction template; paper-specific assignments, file hashes and deterministic indexes accompany it. Auditors retain source access and must correct errors in the brief or packet. Indexes guide inspection and are not evidence.

Each finding states its source location, rerunnable evidence or manual check, confidence (CONFIRMED, LIKELY or SPECULATIVE), benign explanations tested, effect on conclusions and remedy. Tested negatives are retained. Separate workspaces and restricted tools limit information sharing. Preflight and bounded capability probes precede dispatch; promotion requires clean completion, valid structured output, source-hash joins and a closed evidence manifest. Structural repair cannot change scientific content.

3.3 Stage 2: adjudication and drafting

The coordinator reads every area report and records every finding and tested negative in one registry, with disposition INCLUDED, MERGED into a named row, or EXCLUDED with a reason. All rows remain in the ledger. Final finding projections use only eligible, coordinator-approved included rows; removing a disclosure-flagged finding requires a structured evidential basis. The findings appendix, comment metadata, claim map and verification plan are generated from this registry.

A row binds source locations and hashes, the check performed, expected and observed values, tolerances, reviewer evidence, benign explanations and a remedy. Absence of numerical impact alone does not justify exclusion. Projections are regenerated after accepted changes, preserving superseded decisions and wording.

Disputed computations, new material findings and blocking concerns require re-examination. Conclusion-changing claims, coordinator–auditor disagreements and rejected factual criticisms require a structurally different verification route, such as direct calculation alongside author-code execution. A new seed or re-transcribed formula is insufficient. Decisions are ACCEPTED, CONTESTED, WITHDRAWN or UNRESOLVED; accepted corrections propagate to the body, abstract, appendix and verification artifacts. Finding identifiers are fixed before critique.

Three classifications separate consequence, checking effort and disclosure urgency. *Impact* ranges over none, unverifiable, value wrong, conclusion narrowed, conclusion unsupported, conclusion contradicted, and artifact of construction. "None" requires a completed check; "unverifiable" records insufficient evidence. Unsupported is distinct from contradicted, and a result guaranteed by construction can be defective despite exact numerical reproduction.

Verification tier describes human effort: T1 is hand or short-script recomputation; T2 extracts one deposited quantity or executes a bounded pipeline slice; T3 reproduces a full pipeline, table or figure; T4 compares a reproduced result under a specified, defensible alternative analysis; T5 establishes a structural property, potentially through a coun-

terexample or simulation. Tier is independent of impact. A paper-only flag marks findings established from the article alone. *Disclosure* records privacy, consent, ethics or licensing concerns and can accompany any impact (Appendix B).

The isolated first-draft call synthesizes supplied evidence without shell or network access. The surrounding agent session performs executable rechecks. Before cross-family critique, rows remain unapproved and unsettled; these administrative states do not establish that the paper is unaffected.

3.4 Stage 3: adversarial critique

Critics review the audit's claims against its evidence. In round 1, statistics, data, fairness and completeness critics run concurrently; completeness sees all claims. Verdicts are CONFIRMED, REFUTED, OVERSTATED or UNVERIFIABLE, with a blocking flag. A deterministic merge identifies conflicts and high-risk evidence gaps. A targeted adjudicator runs only for a triggered, evidence-complete question; incomplete triggers are deferred for further evidence. The coordinator verifies criticisms before applying or contesting them, and records the integration decisions.

After round 1 is integrated and passing, an included fact enters the settled cache only with coordinator approval and at least two confirmations from distinct reviewers, invocations and evidence records spanning two model families. An auditor supplies the family-A confirmation only for findings it raised; a finding first raised later, by the operator, the coordinator or a critic, goes instead to a blind family-A verifier that sees the claim only as a hypothesis with its source locations. Each claim text receives one verifier verdict (Appendix B). Their evidence hashes are rechecked. Settling a fact reduces repeated review; it does not prove its truth. Later rounds use a hash-bound diff packet. Completeness reviews every changed finding; specialists ordinarily review substantive changes, excluding confirmation and approval bookkeeping. Completeness and one best-matching specialist always run; cross-domain conclusion-changing or disclosure-critical changes can add specialists.

At least two rounds are required so revisions receive a subsequent challenge. Seven is the default ceiling, allowing additional correction–review cycles while bounding repeated inference; it is not a requirement to run seven rounds. Each audit freezes its cap, and a journaled amendment records any increase. `target_complete` requires subsequent review of every changed claim, no unresolved blocking or conclusion-critical item, and no unreviewed changed section. Successful stopping also resolves the last review's refutations, overstatements and blocking items through adjudication or evidenced contests, preserving reviewed substantive claims. `converged` additionally requires a round beyond the target with exactly zero factual corrections. `cap_with_unresolved` and `aborted` record unfinished review with a reason. Failed execution is not a completed round (Appendix B).

3.5 Stage 4: verification and handoff

Release checks bind the report, registry projections, documents, metadata and verification outputs to the current audit state. A notebook declares expected values and justified tolerances before computation and records PASS, FAIL, JUDGEMENT or UNVERIFIABLE. Expectations come from the claim being tested, not the computed observation. Exceptions and unchecked expectations fail; judgment and unverifiability require reasons. The required wrapper must pass and write a current verification manifest.

The numbered handoff connects each finding to a generated R check or an explicit manual/no-code entry. A human verifier records one verdict per claim. Completion of this review is separate from automated release, whose gates validate document bindings, disclosure exclusions and filing receipts (Appendix C). Internal retention and public sharing have distinct disclosure controls. If, after honest study, the verifier cannot establish a claim in either higher handoff category, the verification assignment is returned to the coordinating institution; Appendix B defines the handoff labels and responsibilities.

3.6 Detector controls and disclosure

The *false-clean rule* requires a detector claiming no violations to be tested on known-present and known-absent controls and to report inspected coverage. Scope includes files, sections, lines, pattern forms, encodings and data regimes. Controls must exercise the production path and target form. Appendix G catalogs twenty-two failure mechanisms. Passing controls supports only the tested coverage, not a general guarantee that defects are absent.

A spreadsheet-cell scan cannot establish that document metadata lack identifiers; controls using a different identifier format cannot validate the encountered form. Inspecting no records cannot sup-

port a clean finding about the declared population. The same limits apply to the audit's own outputs. Completion requires checking process status and regenerated artifacts; printed success, partial reports and edited receipts are insufficient.

Personal data in a deposit is a finding whether or not anonymization was claimed. The audit continues with value-free reporting of identifier kinds, locations, affected counts and a verified remedy. Prose redaction preserves computational inputs. The coordinating institution handles disclosure and author or journal correspondence through the filed comment; automated agents and assigned verifiers do not contact them independently.

4 Implementation and Enforcement

The version 2.7.7 implementation packages instructions, policy, runners, schemas and regression checks. Its default family A is Claude Sonnet 5.5 (`sonnet-5-5`, effort `xhigh`) for the five auditors, coordinator and blind verifier. Family B is GPT-6.1 Sol (`gpt-6.1-sol`), at `high` for critics and `max` for triggered adjudication. The baseline uses the same family-A model and effort. Roles remain distinct from model identities; every run freezes its concrete routes and records requested and served identities separately. Appendix D gives per-role limits and supported snapshot behavior.

Three inference starts per role are allowed by default, including retries and eligible repair; critics receive this allowance per round. This bounds recovery from transient failures and invalid outputs without permitting unlimited resampling. Authentication and deterministic capability failures stop dispatch. The five-auditor partition, seven-round cap and three-attempt allowance are inspectable design defaults serving coverage, review and resource control, respectively. Their rationale does not imply numerical optimality.

Invocation lifecycles, grants and gate outcomes enter a locked, append-only, hash-chained journal. Runners record calls and measured environments; the operator records other required events. Metadata and usage summaries derive from this journal. Intake records source and checking-code hashes; later receipts bind executed checks to their inputs. Exact model records, policy, prompts, decisions and stopping evidence distinguish changes in the underlying package from changes in the audit process.

The *contract rule* requires each graded role to receive its validation rules, unless the provider schema prevents invalid outputs by construction. Tests check validator-message coverage and prompt–validator consistency. These checks establish structural validity; scientific requirements such as a defensible benign-explanation test or a distinct verification route require evidence and review.

The default two-round route schedules twelve role executions: five auditors, one coordinator, four first-round critics and two later critics. Access checks, probes, retries, verification and extra routed calls add work and are recorded separately.

5 Baseline and Evaluation Design

The evaluation design pairs the full protocol with an implemented single-agent baseline. Both conditions use the same source package, task definition and finding classifications. Appendix F defines eighteen quantities covering findings, self-correction, compliance, reproducibility, resource use and model sensitivity.

5.1 Single-agent comparison

The baseline uses one family-A agent session to inspect the article and deposit and write the complete comment. It receives the full audit's frozen Phase-1 source files and deterministic indexes, together with the checklist, evidence rules, impact scheme and writing standard. Host-environment provenance is recorded. The agent performs multiple tool turns and can revise its work within the session, including running its output validator. There is no coordinator, external critic, cross-family review or post-session repair. Invalid outputs remain recorded outcomes. Appendix E specifies the implementation.

The default matches the shipped family-A model and reasoning setting, with its own $30 budget and two-hour timeout. Network denial and the shell allowlist follow the auditors' launch contract. Additional restrictions withhold full-audit outputs; a pre-dispatch scan checks readable locations for byte-identical copies. Each run records inputs, condition, environment, validation and observed usage. The conditions have distinct compute budgets: this compares complete workflows, and does not isolate role separation at equal compute.

The baseline uses the protocol's finding classifications, appendix generators and document builder. Its prompt targets the ten-page comment-body cap; the renderer records any excess without a further

Evaluation target	Evidence source
Finding validity	Independent assessment of source evidence, impact and remedies
Protocol compliance (Q15)	Condition-specific contracts, review scope, receipts and output validation
Repeatability (Q16)	Independent repeat audits and blinded matching of eligible findings
Resource use (Q17)	Observed usage, elapsed time, failures and human effort
Model sensitivity (Q18)	Recorded model conditions and a shared finding-matching rule

Table 1: Evidence sources for evaluation quantities.

model session.

5.2 Sampling and assessment

The baseline selector draws from published full audits with retained Phase-1 snapshots. Its default pool matches the shipped model routes; seeded draws stratify by code availability, and explicit audit lists are also supported. A fixed subset manifest records eligibility, selected papers and source hashes. Every attempted package retains a disposition, including invalid output, exhausted budget, timeout and execution failure. Finding estimates use condition-specific eligibility and separate package and finding denominators, with clustering by paper.

The validity criterion is whether a claim follows from its source evidence, has justified impact and identifies a supported remedy. Independent assessment uses the article, deposit and executable checks under a common codebook. Blinded matching distinguishes shared findings from findings unique to either condition. Directional recovery measures agreement; independently established defects supply the reference for missed-defect assessment. Critic verdicts, coordinator decisions and verified comment-defects are separate counting units. Q6–Q8, Q10 and Q11 use the full protocol's review or brief-correction records; absence of critique in the baseline is structural, not a zero-error result.

5.3 Resource accounting and repeatability

Run histories retain successful, failed and superseded attempts. Canonical call records cover both families' observed tokens, inference-started calls and provider-active time (the union of invocation intervals). The paired baseline summary compares family-A tokens and metered dollar cost; family-B calls use the Codex account route, which supplies usage but no attributable per-call monetary charge. Its cost is therefore unavailable, not zero. A family-A subtotal is not a total workflow price.

Budgets and timeouts are reported for both families in Appendix D. Full-pipeline totals include recorded in-audit probes; baseline diagnostics, pre-intake checks, the surrounding orchestration session and human work have separate accounting boundaries. Observed cost and a separately labeled normalization to one-hour cache writes identify the accounting basis. Evaluation retains source-record missingness; summary defaults do not constitute observed zero use.

Q16 compares independent audits under matched initial package, protocol, policy, routing and runtime conditions, with amendments and reopenings recorded. Q18 varies the full protocol's auditor model with the coordinator held fixed. Both use directional denominators and blinded matching. Requested and served identities, inference settings, dates and tool environments identify the comparison conditions. Finding validity, recovery, compliance and resource use remain separate outcomes.

6 Conclusion

The protocol connects language-agent judgments to source evidence, recorded adjudication, cross-family critique and verification before release. Its single-agent baseline preserves the task and evidence standards within one session. The shared finding taxonomy and explicit evaluation quantities connect both workflows to assessments of validity, recovery, compliance and resource use.

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A The Auditor Checklist

Items are numbered for checklist-based coding of findings in the evaluation. Items 1–7 are distilled from published reproducibility-audit comments; items 8–12 were added from calibration and production audits. Table 2 routes the items to areas. Coverage overlaps deliberately, so that each area approaches shared risks from a different evidence path.

Area	Items	Required cross-check
1 Reproduction	3, 8–11	Reproduce every headline value at printed precision
2 Data integrity	1, 2, 4, 6	Trace any anomaly into or out of published statistics
3 Text vs data	3, 5, 9, 12	Derive inputs from the article, not a typed claim list
4 Code, methods	4, 8, 10, 11	Compare code behavior with Methods and dependency versions
5 Privacy, archive	7, 10, 12	Correct the shared brief; verify the live repository record

Table 2: Checklist routing.

1. **Duplication and sample inflation.** Exact duplicates; near-duplicates; contiguous block duplication at every offset; repeat submissions sharing a timestamp or contact; whole-sample duplication. Quantify the coincidence and name the mechanism. Finding no near-duplicates in a survey export does not establish that there are none in the response data.
2. **Impossible and out-of-range values.** Responses outside the scale; subscale scores above the arithmetic maximum; impossible demographics. Trace whether bad values entered published statistics.
3. **Reproduction and source discrepancies.** Sample size in the data versus the paper; means that reproduce while dispersions do not; text versus figure versus data; whole studies missing from an archive that claims completeness. Reproduce at the precision printed and list every value that does not.
4. **Scoring and coding errors.** Subscale formulas mixing opposing-construct items; inconsistent composition across studies; wrong multipliers; silent recoding; scale-offset bugs that manufacture a group difference. Verify the column-to-item mapping against the covariance structure before judging an intended key.
5. **Statistical-reporting anomalies, no data required.** Algebraically impossible statistics; sign errors; missing decimals; mislabeled captions; a wrong direction in an abstract; inconsistently labeled confidence levels; integer requirements rounded down.
6. **Data-distribution anomalies.** Absent latent structure; uniform distributions with sharp cutoffs where skew is expected; impossibly narrow ranges; absurd effect sizes. Benchmark against published reference datasets.
7. **Privacy and archive hygiene.** Personal data is a finding whether or not anonymization was claimed; an unsupported anonymization claim adds a text–deposit discrepancy. Inspect cells, filenames and embedded metadata. Report identifier kinds, file and column locations, affected counts, per-person uniqueness and a verified remedy, never values or re-identification procedures. Use a directory and file count when filenames are identifying. The audit continues with value-free reporting.
8. **Estimator specification and library defaults.** Which function is called, and which arguments are left at their defaults. Print the returned object and check that the element being read is the intended one. Vary the defaults, and check the release notes of pinned dependencies for changed defaults.
9. **Figure fidelity.** Measure plotted values against the text. Axis limits that remove observations differ from coordinate zooms, and a clipped panel silently drops data from its own caption's sample size.
10. **Mechanical reproducibility.** Does the shipped code run as shipped? Look for undeclared dependencies, placeholder paths, runtime fetches by hard-coded identifier, locale-dependent pars-

ing, absent environment capture, and paths that silently load the wrong study's data.

11. **Treatment fidelity, not only outcome provenance.** When a component is cleared as not the source of the dependent variable, ask separately whether it is part of the manipulation.
12. **Reporting omissions.** Computed but unreported analyses. Report them as omissions, not errors, and attribute no motive.

Cross-cutting requirements. Label every claim CONFIRMED, LIKELY or SPECULATIVE. List the benign explanations considered and how each was ruled out. State the impact on published conclusions for every finding; most have none. Report refuted hypotheses as prominently as confirmed ones. Credit what the authors did well, specifically. Verify the remedy, not only the defect: every finding that ends "the authors should do X" asserts that X is possible and sufficient.

B Stage Details

Fan-out runtime. Any tool that may write beside inputs runs on a working copy. A provider-free preflight tests the exact sandbox, read, write and promotion paths; capability probes are bounded and reusable when their bindings match. New snapshots allow 90 minutes per Phase-1 attempt and a $15 cap; supported legacy limits remain recorded per audit. Auditors have no network, plugins or cross-attempt state. Each receives an immutable role packet and read-only deposit access. The closed output directory holds the report, structured result, evidence manifest and listed evidence files. Promotion requires clean completion, schema validity, source-hash joins and manifest closure. Partial output remains diagnostic. The participant–date/time admission rule requires a recognized labeled identifier and a date or clock-time pattern; ordinary participant descriptions and invalid clock ratios do not suffice. Privacy-scan counts and the comment's consistency detector use their own rules. One focused repair may use at most a fifth of the original packet and must preserve scientific fields, report prose and evidence bytes. Live repository checks require externally supplied evidence; unavailable checks remain explicit.

Single-paper and shared deployments. The audit unit is one article and its deposit. One researcher can run an audit. A shared collection adds work allocation and filing. A *reservation* is an advisory claim in a shared directory identifying the operator handling a paper; a competing visible claim is refused. Different papers have separate workspaces and can be audited concurrently. These controls coordinate cooperating clients; they are not a distributed lock service across machines with delayed file synchronization. Per-audit journals use locks, and publication refuses to overwrite an existing destination. The *comments tree* is simply the institution's output directory, organized by journal and paper, with a parallel metadata directory. The institution manages human verification assignments and handles all author and journal correspondence. "Returning a paper" means declining that verification assignment and referring it to the institution; it does not mean contacting the paper's authors.

Access and snapshot checks. Intake makes or reuses a minimal critic-access call before reservation or package copying. A successful check is cached for 24 hours and bound to the host CLI, account state and route; failures are not reused as passes. It is separate from an audit's capability probes. Dispatch verifies frozen input hashes and inventory before provider calls. Harmless permission-mode drift can be repaired only after content integrity passes; content changes require rebuilding the appropriate state rather than accepting a different snapshot.

Coordinator isolation. The isolated first-draft call reads an immutable snapshot and writes only its output directory, with neither network nor shell. Preparation joins all five canonical reports to completed auditor invocations, constructs draft registry rows, and builds privacy-reviewed evidence with manifest closure. The call synthesizes that evidence without executable re-runs. Admission checks include route-matching served-model identity and exact input, output and normalized-total token counts, schema-valid findings and evidence joins. Auditor and verifier outputs and coordinator inputs share a participant-code admission check; code-shaped variable names receive only evidence-based exemptions. The first draft, adjudication log, registry and retired-claims file are promoted together; the runner also creates the working report transactionally. Failed attempts preserve canonical files. Executable re-runs and second routes occur outside the call, followed by critique integration.

Registry and decisions. A finding row binds stable identity, source locations and hashes, opera-

tions, expected and observed values, tolerances, benign explanations, dependencies and disposition. Final projections select approved included rows; the ledger retains every row. Accepted, contested, withdrawn and unresolved decisions retain their history. Re-applied integrations can mark superseded decision blocks withdrawn without withdrawing the finding. Removing a disclosure-flagged finding requires refutation, duplication into a final finding, absence of a disclosure defect, or a tested negative. Successful releases cannot leave included rows unapproved or nonfinal; unfinished stops may explicitly retain unresolved disclosure findings. These disclosure checks are blocking for audits initialized or reopened under version 2.6.9 onward; older audits receive warnings.

Classification. Each finding is classified on three separate, protocol-defined fields. Editorial guidance informs the treatment of reliability and corrections (COPE Council, 2019; International Committee of Medical Journal Editors, 2026); NISO CREC concerns communicating editorial status (National Information Standards Organization, 2024). These sources do not define the protocol's ordered impact taxonomy.

- *Impact*, on an ordered ladder: *none*, *unverifiable*, *value wrong*, *conclusion narrowed*, *conclusion unsupported*, *conclusion contradicted*, and *artifact of construction* (a result guaranteed by how the analysis was built). *None* requires that the check was possible and completed; a check the package could not support is *unverifiable*.
- *Verification tier*: T1 recomputes one or two quantities by hand or in a few lines; T2 extracts one deposited quantity or runs a bounded pipeline slice; T3 rebuilds the environment and reproduces a full pipeline, table or figure; T4 reproduces and compares a specified, defensible alternative analysis; T5 establishes a structural property, such as a result entailed by construction or a validation absent throughout the analysis pipeline, potentially with a counterexample or simulation. A separate flag marks findings established from the article alone. Tier measures human verification effort, never gravity. For an unverifiable finding, tier measures the work needed to establish the evidentiary gap; the blocked check's would-be tier is recorded separately.
- *Disclosure*: privacy, consent, ethics and license flags with an urgency. It can co-occur with any impact, including *none*.

These labels organize findings; editorial action remains a separate judgment. At comment level, impact is the maximum individual impact, never a sum of lesser defects, and residual unverifiability is retained separately. The protocol does not recommend editorial action or assert misconduct or intent.

Critic packets. Round 1 routes quantitative and methods claims to statistics; provenance, counts, exclusions and duplicates to data; disclosure, archive, attribution and framing to fairness; and all claims plus report-wide joins to completeness. The driver freezes the report reviewed. Later packets carry changed findings, sections, dependencies, settled references and routing signals. The settled set binds the current report and registry; later refutations or overstatements can invalidate earlier confirmations even when the later round is not integrated. A verdict is distinct from confirming evidence: an overstated claim may be confirmed after correction, while an unverifiable verdict may support a claim that verification is impossible. Confirmations must support the integrated wording.

Later-round routing. Fairness covers disclosure, remedy, impact, attribution and proportionality; statistics covers quantitative, estimator and inference changes; data covers sample, exclusions, missingness, records and manifests. New snapshots use substantive-change routing for specialists: added or removed findings, substantive registry fields and changed finding-section text enter scope; confirmation and approval bookkeeping alone does not. Completeness still reviews every changed finding. Older snapshots retain their routing rule. With no substantive specialist scope, completeness and the first routed specialist remain, with that specialist reviewing all changed claims. A no-change packet defaults to fairness plus completeness. No-change rounds require generated, content-bound packets.

Post-fan-out verifier. The verifier runs only on included rows that no Phase-1 auditor raised or independently found, and only for claims that cite a deposited file at the Phase-1 snapshot hash. Each call is recorded at round 0 in family A on the auditors' frozen route, reusing their capability probe. It has one attempt budget per batch of claims and runs under the fan-out sandbox, admission and single-repair contract. Its packet carries the claim text, source location and snapshot hashes only; critic verdicts, coordinator reasoning, registry status and

finding identifiers are withheld, and each attempt is denied read access to earlier attempts, other roles' outputs and late release artifacts. It returns confirmed, refuted or unverifiable per claim. A verdict binds to the SHA-256 of the current claim text, so renumbering cannot misbind it and an edit voids it. A refuted or unverifiable verdict blocks any verifier confirmation of that claim text, including after its record is superseded. Integration binds only calls that ended no later than its timestamp, and every downstream check re-admits a record through the runner's own validation, so a hand-assembled record is refused.

Stopping evidence. Every stop records the union of changed claim identifiers, subsequent review, unresolved blocking and conclusion-critical items, and unreviewed sections. Successful stops require the unresolved lists empty and preserve the last-reviewed substantive report and findings. The last round's refutations, overstatements and blocking items must be resolved by the relevant adjudication or an explicit contest stating both position and evidence in `CONTESTED.json`; convergence additionally checks other requested corrections. Contests must be staged with the release. `converged` also requires exactly zero factual corrections in a round beyond the target. A terminal round with no factual correction normally records verdicts and stopping evidence without integration; `target_complete` permits limited integration for deferred approvals or contests followed by regeneration. At the effective cap, unresolved review is recorded as `cap_with_unresolved`; a genuine blocker or failed run uses `aborted`, both with an explicit reason.

Retries and grants. The default budget is three inference starts per role, across retries, repair and resumed execution; critics receive a separate budget for each round. The ceiling may be raised to twenty. Transport and transient provider failures may retry. Phase-1 and coordinator output overflows permit at most one fresh automatic retry. Phase-1 and coordinator timeouts stop the invocation; critic and targeted-adjudicator wall-clock timeouts can retry within budget. Invalid sampled outputs can consume the remaining budget; eligible structural repair does not permit scientific changes. Preflight and rejected schema admission do not consume inference starts, although repeated admission requests have a separate bound. The coordinator charges timeouts even without a result envelope; Phase 1 uses reported inference or new workspace files as evidence that inference began. Authentication, authorization, quota, unsupported-model, capability and deterministic provider-schema admission failures stop the invocation. Each round's immutable plan pins roles, routes, efforts and data classes and receives a hash-bound journal grant before dispatch; attended mode requires the exact plan to be authorized.

Verification notebook. The notebook is written for a verifier who did not run the audit and is not expected to trust it. JUDGEMENT marks a claim resting on an assessment, with its checkable substrate checked; UNVERIFIABLE carries its reason. Expected values are declared before observed values are computed, with justified tolerances. Declared but unchecked expectations fail. The wrapper renders the notebook, requires a passing status, and binds the result by hash to the notebook, report, comment record and package manifest. A separate snippet runner can exercise checks in the notebook's setup and prelude before release; it produces neither verification status nor manifest and does not replace the wrapper.

Claims script and filing. The canonical projection maps executable checks to R snippets and other findings to a no-code tail. The standalone builder supports verbatim author-code mappings, but the canonical projection does not emit them; hand-changing the map fails the release projection check. Findings verified by inspection or arithmetic go in a closing no-code section, each with a one-line manual check. The script maps impact and confidence to retraction-, corrigendum- or comment-material labels; a claim that remains unconfirmable after honest study in either higher category requires the verifier to return the verification assignment to the coordinating institution and take another paper, rather than omit the claim or delegate its verification to an AI. These are handoff labels, not recommendations to editors. Speculative findings receive comment material regardless of impact. Among the remaining findings, confirmed unsupported or contradicted conclusions receive retraction material. The same impacts at likely confidence, and value-wrong, conclusion-narrowed or artifact-of-construction findings, receive corrigendum material; other findings receive comment material. Each verifier records one verdict per numbered claim, including those in the no-code section. At publication the PDF and DOCX go to a per-paper folder in the

comments tree; the claims script, metadata pair, finalization receipt and manifests go to a matching metadata folder. The release manifest is written last, under a lock. This is internal filing: run and comment metadata are marked unsafe for direct public sharing and require their declared redactions.

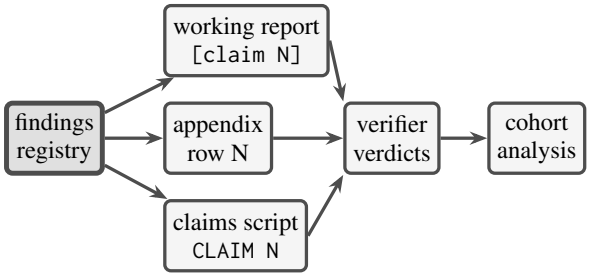


Figure 2: The handoff chain. The registry generates three numbered artifacts, and the verifier's per-claim verdicts join back on the claim number.

C The Release Gates

Release policy requires blocking checks to fail closed. Implemented blocking failures return nonzero and prevent release; some checks issue advisory warnings. The consistency gate runs its four blocks and accumulates their blocking failures. These checks have finite coverage and do not certify every semantic requirement.

Consistency gate. This gate applies four groups of configured checks to the comment.

1. The disclosure sweep tests participant-code patterns on planted examples, including an underscore-prefixed form, and blocks recognized same-line code–date or code–clock matches.
2. Supplied retired phrases are matched against individual lines throughout the document after normalization. Summary numbers missing from the body produce warnings.
3. House checks enforce the abstract word limit and absence of tables, required citation markers, numbered-table placement and a Recommendations-heading prohibition. They flag configured process-narration phrases; a legacy Executive summary heading is accepted with a warning, although the current document specification requires an Abstract.
4. Planted examples test the appendix-row and claim-tag extractors, which then check identifier uniqueness, matching order and body-section references.

Post-delta gates. Before each later round, consistency, release-bound privacy, canonical projections and archive integrity run concurrently and produce a passing content-bound receipt. Some checks use copied inputs; others use the live audit root with checked bindings. Ordinary later-round dispatch with canonical projections refuses missing, failed or stale receipts; a same-input adjudicator-only retry is excepted. Finalization replays required receipts, verifies that the last round's passing gate was journaled before review began, and reruns the latest bound checks. A receipt generated after review does not establish prior gating.

Document build. The build binds the report, front matter, build code and tool identities to an atomic DOCX/PDF pair. Its ten-page comment-body cap comes from the audit's policy snapshot; preview caps do not authorize release. References and appendices begin on separate pages. Rendered findings-table checks preserve literal source text in DOCX cells and detect fragmented words in the PDF, with documented exceptions for overlong tokens. Release boundaries recount the exact PDF; a failed build preserves the previous passing pair.

Verification. The notebook must render and pass, and its manifest must bind it to the current report, comment record and package manifest. Judgment and unverifiable claims must state why.

Staging. Staging validates mandatory inputs and document bindings, removes verbatim deposit copies and normalizes machine paths. It blocks prohibited data-file extensions, except the required verification summary, and configured identifier–timestamp matches. Participant-list checks require eight matching lines and 25% density; opaque-key checks require twenty distinct matching tokens. Supported text/code and extracted PDF/DOCX content are inspected, with finite coverage. An additional check compares release outputs with identifying values from supported deposit fields using temporary hashes; unreadable inputs are reported as unscanned. Staged copies of retained records can mask matched values, home-account segments and disallowed email addresses, with value-free reports of locations and counts. Canonical audit records remain unchanged; identifying content in the comment's own outputs still blocks release. Package-specific checks and the PDF recount also apply.

Finalization. Finalization first self-tests its schema validator against known-good and known-bad documents, and refuses to run if the validator fails to reject the bad ones. It then derives the computed fields of the run and comment records, validates both, cross-checks their identifiers, and files them into the program registry with a receipt. The stop value must be supplied. It is a fact about the run, not something a script infers. Finalization refuses the skill-local registry and prints the registry it files into.

Journal privacy and continuation. Operator-supplied free text is checked before append. A redaction event can mask a named field of an earlier event in projections while leaving the original journal intact. Masked events in the run projection are checked through hash-chain continuity rather than re-hashing altered text; original journal events remain fully hash-checked. A consistency gate logged at round 0 stays in the compliance record without creating a critique-round row. Environment capture is a measured, tool-generated journal event. Finalization closes the audit to further critique until a controlled reopening. Reasoned cap amendments and reopenings preserve superseded snapshots and receipts and are revalidated before finalization and publication; they are not manual snapshot edits.

Publication. Publication requires a current registry-filed receipt and intact bindings for finalized artifacts, including reviewed-state and disclosure checks. Permitted post-finalization journal events are checked against the preserved chain prefix; other bound changes invalidate the receipt. Existing destinations are never overwritten.

D Default Policy

Table 3 specifies the shipped models, reasoning effort and limits. The family-A route is `sonnet-5-5` (spawn alias `claude-sonnet-5-5`); family B is `gpt-6.1-sol`. These are requested routes; the journal records served identity separately. New audits use the shipped configuration and critic-route lock. Account incompatibility stops dispatch. Supported legacy snapshots retain their recorded settings and any authorized migration history. Deterministic merging and release gates add no model session; intake may make a minimal critic-access call. Surrounding orchestration is distinct from the isolated drafting call.

The default route runs five auditors in parallel, four critics in round 1, then completeness and one routed specialist; triggered cross-domain review adds specialists. The blind verifier and targeted adjudicator run only when their conditions apply. Two rounds are required, with a frozen new-audit cap of seven. The cap was raised from five after audits reached that boundary with unreviewed changes; it remains a resource ceiling rather than a confidence threshold. Three inference starts per role include retry and repair (per round for critics); the configurable ceiling is twenty. Settling a fact requires coordinator approval and two separate confirmations spanning both families after round 1.

Role	Model / effort	Per-call limit
Five auditors	Sonnet 5.5 / xhigh	90 min; $15
Draft coordinator	Sonnet 5.5 / xhigh	120 min; $20
Blind verifier	Sonnet 5.5 / xhigh	90 min; $15
Critics	GPT-6.1 Sol / high	90 min
Targeted adjudicator	GPT-6.1 Sol / max	90 min
Single-agent baseline	Sonnet 5.5 / xhigh	120 min; $30

Table 3: Default model routes and runner limits. Family-B calls have no explicit dollar or total-token cap in the launch contract; time and attempts bound them. Family-A dollar caps apply per attempt and can be exceeded by the turn in progress.

The Sonnet 5.5 route requests a 128,000-token *per-response output cap*, not a total session-token budget. Ordinary Phase-1 and critique probes use their respective role routes. The separate baseline diagnostic probe uses Sonnet 5.5 at low effort, a $1 budget and a ten-minute timeout. Probes test the output contract; missing provider identity remains missing. A drift check compares maintained skill documentation and runtime contracts with policy.

Resource records retain provider-specific token semantics. Family-A totals include fresh input, cache-read input, cache-created input and output. For family B, input plus output gives the total when a direct total is absent and both components are observed; cached input and reasoning are subsets and are not added again. Summaries retain observation coverage and separate summed call duration from the union of active intervals. Family-A metered cost is observed; the family-B adapter records monetary cost as unavailable. The baseline comparison utility selects family-A token and cost subtotals, while its call and active-time measures span both families.

E Single-Agent Baseline

The baseline implementation is version 1.1.0, with task references copied from protocol version 2.7.7. It defines a separate single-agent, single-pass condition: one model session covers all five audit areas and writes the final comment. Self-checking and tool use occur within that session; no later model call repairs invalid scientific output.

Sources and sampling. The selector reads published full audits with retained Phase-1 snapshots. Seeded draws can preserve the pool's code/no-code proportions or balance those strata; explicit audit lists define a separate selection route. The subset manifest records the selection method, pool membership, hashes and source audits. Replacing a used subset preserves its predecessor. Raw article and deposit files and deterministic indexes are copied from the selected source snapshot and checked against its manifest, except that the environment fingerprint is regenerated when the shared helper is available; otherwise it is labeled as describing the source host. Full-audit findings, critiques and the coordinator's working brief are excluded.

Task and inference contract. The agent receives the checklist, evidence standard, classification scheme, report structure, citation rules and baseline findings format. Process instructions about critique, adjudication and ledgers do not apply to this condition. The prompt requires benign-explanation checks, tested negatives, verification of remedies and a structurally different route for conclusion-changing findings. It supplies the output validator for self-checking within the session. The default matches the shipped family-A route and effort, expects a one-million-token context window, requests a 128,000-token per-response output cap and one-hour prompt-cache writes, and sets a $30 call budget and 7,200-second timeout. Observed spending includes any overshoot by the turn in progress. Served identities and reported windows are checked; a second served model or mismatched window is a condition error, and missing window telemetry stays missing. The capability probe checks the requested output cap from provider request logs when those logs are available. Cache-write lifetimes and their coverage are recorded.

Access and contamination controls. The launch follows the auditors' shell allowlist and network denial while removing parent-session messaging credentials and identifiers. File tools are restricted to the run's input, packet and output directories and saved tool output. Shell denial covers the shared project tree, cloud folders, common personal-data locations, agent histories and other baseline runs. Before paid dispatch, the runner searches the permitted scan roots for byte-identical copies of the selected papers' completed comments and metadata and refuses a match. This check records its searched roots and coverage; it tests exact-copy contamination. The host toolchain and environment are recorded separately from the original full audit.

Outputs and outcomes. The agent writes front matter, a final report, captions, a structured findings record and supporting evidence. Shared generator functions derive the findings appendix and cross-reference, and the full protocol's builder produces PDF and DOCX. The baseline renderer lifts the enforced body-page cap, records the page count and flags whether the output exceeds ten pages; rebuilding documents changes no model finding. Successful calls with invalid outputs remain `completed_invalid`; budget exhaustion, timeout, failure, interruption and abandonment have distinct statuses. A transient or transport error permits one extra call only when neither inference nor output files indicate that work began. A rerun has a new run identifier and budget, with earlier outputs retained as superseded records. Calls execute serially under a batch lock.

Retention and accounting. Every run contributes to the run log. Hygiene screening withholds flagged outputs, unreadable text or PDFs, and unsupported data formats from the shared tree, recording each withheld file and retaining originals privately. Input hashes, code identity, condition, environment, call records, validation and rendering history connect each output to its run. The resource summary includes all runs under the current condition, including superseded and failed attempts. Full-pipeline call counts and provider-active elapsed time cover both families; the paired token and cost comparison covers family A. Full-pipeline totals include recorded in-audit probes; baseline diagnostic probes and pre-intake access checks have separate attribution. Summary groups are keyed by model, effort, budget, timeout, prompt-template hash and cache lifetime; CLI and other caching settings remain recorded comparison fields. Observed cost is distinguished from cache-

normalized cost, which reprices cache writes at the configured one-hour rate. Unknown inference-started-call costs make baseline aggregate cost unavailable. Token totals sum available entries, and absent full-pipeline active-time totals default to zero in the summary; source records retain the corresponding missingness.

F Evaluation Quantities

Retained artifacts. Each full-protocol audit keeps the comment and its build manifest; the five auditor reports, the registry and its generated projections, the adjudication log and the retired-claims file; every critic output, response and contest record, the settled-fact cache, diff packets and post-delta receipts; the verification notebook with its manifest, and the claims script; and the journal, redaction records, policy snapshots and amendment history, projections, and run and comment records validated against published schemas. The full audit record is retained internally, distinct from the bounded filed artifact set. Process quantities use these artifacts; independent ratings, repeat runs, model comparisons and remediation follow-up use their corresponding evaluation records. Public sharing remains subject to disclosure controls.

Table 4 defines evaluation quantities without amending the skill. Full-protocol eligibility selects included, coordinator-approved substantive findings. Baseline findings use the corresponding fields in their own structured records, with validation status retained and a common assessment codebook; they are not assigned coordinator approval or cross-family confirmation. Tested negatives and merged duplicates are excluded from defect shares. Q6–Q8, Q10 and Q11 use full-protocol review or brief-correction records; Q18 varies its auditor model. Compliance is assessed against each condition's own contract. Q13 additionally retains unresolved disclosure flags at unfinished stops, reported separately from supported exposures. Finding outcomes may use eligible completed audits; stopping, archive-availability and compliance summaries retain all applicable sampled packages. Analysis specifications fix sampling, eligibility, tokenization, severity ties, denominators, deduplication, blinded matching and endpoints. Ratios with zero denominators are undefined, with their counts retained. Recovery measures agreement, not accuracy or missed-defect recall.

#	Quantity	Operational definition and unit	Estimation
Q1	Conclusion-changing share	At least one eligible finding has `changes_conclusion=true`; the affected conclusion is recorded separately and impact is independently assessed. Unit: paper	Proportion with interval; all dispositions reported
Q2	Reproducible-but-wrong share	Headline quantities recovered to printed precision and Q1 satisfied. Unit: paper	Joint proportion with interval; reproduction $\times$ conclusion-change counts
Q3	Defect taxonomy shares	Impact and study-coded locus among eligible substantive findings. Unit: finding, clustered by paper	Cluster-bootstrap intervals
Q4	Single-token cause share	Most severe finding attributable to one lexical token in code or data, coded by two raters. Unit: paper	Proportion with sensitivity range
Q5	Single-token remedy share	Changing that token and re-running unchanged code suffices: no new estimand, uncertainty computation or author decision. Unit: paper	Reported with Q4, never merged
Q6	Blocking comment-defect count	Adjudicated comment-defects among claims or new issues marked blocking in round r; correction status recorded separately. Unit: round	Counts by round and pooled
Q7	Contested-criticism share	Criticisms contested rather than applied; among contested items, the share later independently upheld. Unit: criticism	Both numerators and denominators, by round
Q8	Substantive withdrawal share	Logical adjudication decisions withdrawing a finding or claim, deduplicated across re-applied integrations; administrative supersession is separate. Unit: decision	Denominator: all deduplicated logical decisions; critic-induced subset separately
Q9	Detected false-clean incidence	Distinct clean statements later disproved, classified using Appendix G. Denominator: deduplicated clean statements. Unit: statement	Detected incidence; follow-up and control coverage
Q10	Brief-correction rate	Auditor corrections to the brief or role packet, from each report's brief-corrections section, tiered central, material or incidental. Unit: audit	Share with a central correction, incentive disclosed
Q11	Stopping outcome	Stop value, rounds, effective cap, amendments and reopening segments; first qualifying convergence and unreviewed sections. Unit: audit	Shares by stop and cap; later convergence unobserved
Q12	Unconfirmed-concern density	Tested candidate concerns not confirmed, deduplicated by the codebook, relative to substantive findings. Unit: comment	Counts and ratio; undefined if denominator is zero
Q13	Disclosure incidence and lag	Supported exposures and separate unresolved flags; handoff, observed notification and verified remedy. Unit: audit (incidence); supported exposure (lag)	Censor lag only with observed origin; unconfirmed flags indeterminate
Q14	Archive-defect incidence	No data; no code; execution failure; no environment record; broken pointer; no license. Unit: package	Applicable proportions; unassessable checks separate
Q15	Protocol compliance	Condition-specific requirements: full-protocol snapshots, review and release bindings; baseline input, model, output-validation and render records. Unit: audit	Compliance vector; all dispositions reported
Q16	Protocol reproducibility	Repeat audits with matched initial package, protocol, policy, routing and runtime; blinded finding matching; amendment/reopening handling prespecified. Unit: finding, clustered by package	Directional recovery and new findings; clustered intervals
Q17	Cost	Observed usage and cost with model-family coverage; calls, failures, pre-intake checks, active time and human effort; reopening gaps separate; unknowns missing. Unit: run, phase or round	Observed resource use; coverage and missingness
Q18	Auditor-model sensitivity	Full-protocol alternative auditor models with fixed task, prompts and coordinator; report changed runtime configuration. Unit: finding, clustered by package	Directional recovery and new findings by model and area

Table 4: Evaluation quantities, counting units and estimation rules.

G Twenty-Two False-Clean Mechanisms

The entries are failure modes to check for and carry no incidence counts. They are the recognizable ones from software-quality practice: inadequate input coverage, boundary and encoding defects, fail-open behavior, tests that exercise the wrong path, stale fixtures and self-certification. Each is named generically so that a comment can cite it by number. Several are created by the audit rather than inherited from the paper, because safeguards compose badly. The design response is to check completion signals independently of the detector and treat missing completion evidence as a failure rather than a pass.

#	Mechanism	Illustration
M1	The pattern cannot express the target's format	A birth-year pattern run against full dates; an identifier pattern that assumes one naming convention
M2	A word-boundary anchor never fires	A \b anchor before a token preceded by an underscore, which is itself a word character
M3	Encoding scope wrong	UTF-16 strings invisible to an 8-bit pattern; a byte-order mark silently renaming a column; document formats never decoded although they ship
M4	File scope wrong	A scan covering one subdirectory while the claim covered the staged tree; a suffix allow-list that omits the data formats
M5	Section scope wrong	A retired-claims check that reads the abstract and not the body
M6	A conjunction too strict	A matcher requiring an identifier immediately before a timestamp misses rows with an intervening measurement
M7	A gate prints FAIL and exits zero	The failure is visible in the log but release proceeds; exposed by planting a violation and checking the exit status
M8	A silent skip reported as a pass	A missing claims file, an undelimitable abstract, an empty keyword or an absent draft, each treated as nothing to check
M9	A positive control validates the wrong code path	The control exercises one matcher while the detector uses another
M10	A positive control validated against the wrong form	Planted year-bearing dates certify a detector blind to day-month dates
M11	A positive control trips the gate it validates	Mitigated by splitting literals and using out-of-range values; the mitigation itself can become dead code
M12	A superseded revision reviewed or staged	A critique round reviews a draft already revised; a stale report is staged silently
M13	A stale prompt produces confident, irrelevant output	A critic reviews a different paper entirely; the failure surfaces as reassurance, not as an error
M14	A detector run outside its preconditions	Granularity tests on non-integer inputs; a duplicate-detection null destroyed by legitimate block missingness
M15	A detector with no power	A dispersion screen that accepts nearly the whole candidate space at the relevant subgroup sizes; an underpowered digit test
M16	Vacuous input	A scan over empty rows; a correction keyed to a participant code that does not exist; a range check on a column coerced to text
M17	A wrong denominator	Re-identification risk divided by distinct key combinations rather than by released records
M18	An arithmetic reconciliation that closes for the wrong reason	A total that balances under two different explanations, only one of which was checked
M19	A seed that does not pin the result	A fixed split seed operating on a hash-ordered set; resampling with no seed call at all
M20	A guard that masks the defect on the production path	A length guard writes missing values only on full-length runs, so printed tables look clean while saved predictions are corrupt
M21	Verifying the wrong dimension	Every cell verified against its cited source line while the error is in the column heading
M22	The integrity mechanism certifies itself	A ledger self-audit that tracks only findings already in the draft, and so cannot detect a finding that never reached it

Table 5: False-clean mechanisms.